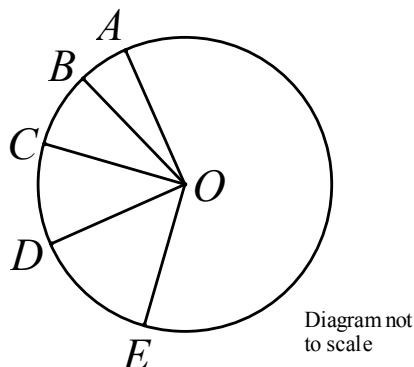


Question 1.

Marks

- (a) Find $\lim_{x \rightarrow 0} \left(\frac{\sin 2x}{\tan 5x} \right)$ 2
- (b) If $x = \alpha$, $x = \beta$ and $x = \gamma$ are the roots of the equation $3x^3 + 4x^2 - 2x + 1 = 0$, find the value of $\alpha^2 + \beta^2 + \gamma^2$. 2
- (c) Find the angle to the nearest minute between the lines $4x - 5y + 3 = 0$ and $y = 3x - 2$. 2
- (d) Circle with centre O and radii AO, BO, CO, DO and EO is shown below with $\angle AOB = 15^\circ$, $\angle BOC = 25^\circ$, $\angle COD = 35^\circ$ and $\angle DOE = 40^\circ$. Prove that $AD = CE$ 3



- (e) (i) Show that $x - 3$ is a factor of $x^3 - 12x^2 + 47x - 60$. 1
 (ii) Hence express $x^3 - 12x^2 + 47x - 60$ as a product of linear factors. 2
- (f) Find an expression between x and y that is independent of t when $x = e^t + e^{-t}$ and $y = e^t - e^{-t}$. 2
- (g) Find the number of ways all the letters from the word *choice* can be arranged around a circle. 1

Question 2.

- (a) If $\sin A = \frac{1}{4}$ and angle A is obtuse, find the exact values of : 2
 (i) $\sin 2A$ 2
 (ii) $\cos \frac{A}{2}$ 2
- (b) A polynomial $P(x)$ has a remainder of 2 when divided by $x - 1$ and a remainder of 3 when divided by $x + 2$. 3
 Find the remainder when $P(x)$ is divided by $x^2 + x - 2$.
- (c) How many ways can a group of 2 men and 3 women be chosen from 6 men and 5 women? 2
- (d) (i) Write $\sqrt{2} \cos x + \sqrt{7} \sin x$ in the form $R \cos(x - \alpha)$, where $R > 0$ and α is to the nearest minute. 2
 (ii) Graph $y = \sqrt{2} \cos x + \sqrt{7} \sin x$ for $0^\circ \leq x \leq 360^\circ$, showing all points where the curve crosses the x and y axes. 2
 (iii) Solve $\sqrt{2} \cos x + \sqrt{7} \sin x = 1$ for x in the domain $0^\circ \leq x \leq 360^\circ$ 2

Question 3.

- (a) Find the number of words that can be formed using all the letters from $\{a, b, c, d, e, f, g\}$ if a vowel is at each end and the letters f and g are together. 2
- (b) Find the equation in expanded form of a monic cubic polynomial with real roots : 2
 $1, 2 + \sqrt{3}$ and $2 - \sqrt{3}$.
- (c) Differentiate with respect to x in simplest terms:
- (i) $y = \frac{\sin(x+7)}{\cos(x-9)}$ 3
- (ii) $y = \log_x 3$ 2
- (d) The points $P(2p, p^2)$ and $Q(2q, q^2)$ lie on the parabola with equation $x^2 = 4y$, such that $pq = 4$. The normals to the parabola at P and Q intersect at N .
- (i) Derive the equation of the normal at the point P . 2
- (ii) Show that the intersection point N of the normals is $(-4(p+q), p^2 + q^2 + 6)$. 2
- (iii) Find the Cartesian equation of the locus of N , stating any restrictions. 2

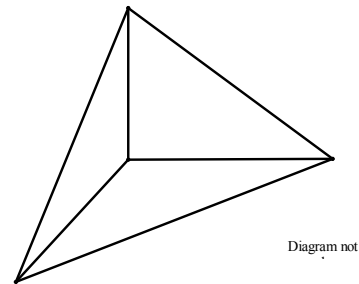
Question 4.

- (a) A vertical post BD of length h metres is held in position by ropes AD and DC . The distance AC is 30 metres, $\angle ABC = 100^\circ$ and the angles of elevation of D from A and C are 10° and 8° respectively.

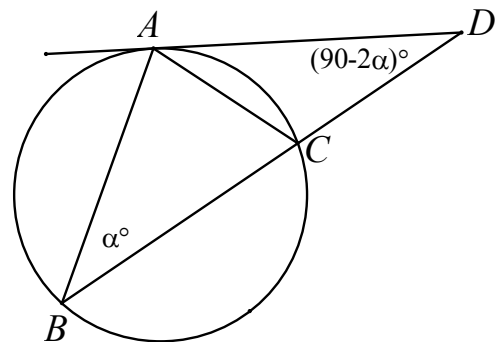
- (i) Show that

$$h = \frac{30 \tan 80^\circ \tan 10^\circ}{\sqrt{\tan^2 8^\circ + \tan^2 10^\circ - 2 \tan 8^\circ \tan 10^\circ \cos 100^\circ}}$$

- (ii) Find the height of the post to the nearest centimetre.



- (b) Triangle ABC is inscribed in a circle. The tangent to the circle at A meets BC produced at D as shown with $\angle ABC = \alpha^\circ$ and $\angle ADC = (90 - 2\alpha)^\circ$, where $0^\circ < \alpha^\circ < 45^\circ$. Prove BC is the diameter of the circle.



- (c) (i) Show $\frac{d}{dx} \left(\frac{x^2 - 3x + 2}{x^2 + 4x + 3} \right) = \frac{7x^2 + 2x - 17}{(x^2 + 4x + 3)^2}$ 2
- (ii) Find the location of the turning points of $y = \frac{x^2 - 3x + 2}{x^2 + 4x + 3}$, and test their nature. 3
- (iii) Graph $y = \frac{x^2 - 3x + 2}{x^2 + 4x + 3}$, showing all asymptotes, turning points, and points where the curve crosses the coordinate axes. 3

End of Exam

$$(a) \lim_{x \rightarrow 0} \frac{\sin 2x}{\tan 5x} = \lim_{x \rightarrow 0} \left[\frac{\sin 2x}{x} \cdot \frac{5x}{\tan 5x} \cdot \frac{2}{5} \right]$$

$$= 1 \cdot 1 \cdot \frac{2}{5}$$

$$= \frac{2}{5}$$

$$(b) \angle APB = -\frac{4}{3}$$

$$\angle BAP + \angle BPA = -\frac{2}{3}$$

$$\angle PAB + \angle PBA = (\angle APB)^2 - 2[\angle PAB + \angle PBA]$$

$$= \left(-\frac{4}{3}\right)^2 - 2 \cdot \left(-\frac{2}{3}\right)$$

$$= \frac{16}{9} + \frac{4}{3}$$

$$= \frac{28}{9}$$

$$= 3\frac{1}{9}$$

$$(c) m_1 = \frac{4}{5}, m_2 = 3$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{\frac{4}{5} - 3}{1 + \frac{4}{5} \cdot 3} \right|$$

$$= \left| \frac{-\frac{11}{5}}{\frac{13}{5}} \right|$$

$$= \frac{11}{13}$$

$$\theta = 32^\circ 54'$$

$$(d) \angle AOB = 15 + 25 + 35$$

$$= 75^\circ$$

$$\angle COE = 35 + 40$$

$$= 75^\circ$$

$\therefore AD = CE$ (Equal angles subtend equal chords)

$$(e) (i) P(3) = 3^3 - 12 \times 3^2 + 47 \times 3 - 60$$

$$= 27 - 108 + 141 - 60$$

$$= 165 - 168$$

$$= -3$$

$\Rightarrow x-3$ is a factor

$$(ii) x^3 - 12x^2 + 47x - 60 = (x-3)(x^2 - 9x + 20)$$

$$= (x-3)(x-4)(x-5)$$

$$(f) x^2 = x^{2t} + 2 + x^{-2t}$$

$$y^2 = x^{2t} - 2 + x^{-2t}$$

$$\therefore x^2 - y^2 = 4$$

$$(g) \begin{matrix} c & c & c \\ & 4 & \\ c & & 0 \end{matrix} \text{ ways} = \frac{5!}{2!}$$

$$= 60$$

①

2(a)

$$(i) \sin 2A = 2 \sin A \cos A$$

$$= 2 \cdot \frac{1}{4} \cdot \frac{-\sqrt{15}}{4}$$

$$= -\frac{\sqrt{15}}{8}$$

$\cos A < 0$
A obtuse

$$(ii) \cos A = 2 \cos^2 \frac{A}{2} - 1$$

$$2 \cos^2 \frac{A}{2} = 1 + \left(-\frac{\sqrt{15}}{8}\right)$$

$$\cos^2 \frac{A}{2} = \frac{8 - \sqrt{15}}{16}$$

$$\cos \frac{A}{2} = \frac{\sqrt{8 - \sqrt{15}}}{4}$$

Half obtuse $\rightarrow$ acute angle
 $\cos \frac{A}{2} > 0$

$$(h) P(x) = (x-1)(x+2)Q(x) + ax + b$$

$$x=1 \quad 2 = a+b$$

$$x=-2 \quad 3 = -2a+b$$

$$a = -\frac{1}{3} \quad b = 2\frac{1}{3}$$

$$\text{Remainder } R = -\frac{x}{3} + 2\frac{1}{3}$$

$$(c) \text{No ways} = \binom{6}{2} \binom{5}{3}$$

$$= 15 \times 10$$

$$= 150$$

$$d(i) R_{\cos}(x-d) = R_{\cos x \cos d} + R_{\sin x \sin d}$$

$$R_{\sin d} = \sqrt{7}$$

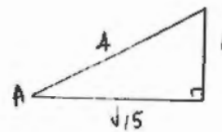
$$R_{\cos d} = \sqrt{2}$$

$$\tan d = \sqrt{\frac{7}{2}}$$

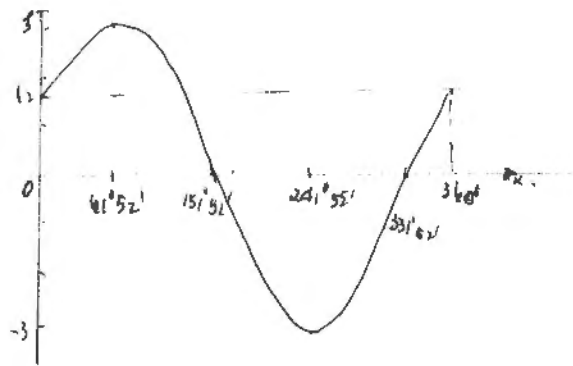
$$d = 61^\circ 52'$$

$$R = 3$$

$$\therefore \sqrt{2} \cos x + \sqrt{7} \sin x = 3 \cos(x - 61^\circ 52')$$



②



(iii) $\sqrt{2} \cos x + \sqrt{7} \sin x = 1$

$$3 \cos(x - 61^\circ 52') = 1$$

$$\cos(x - 61^\circ 52') = \frac{1}{3}$$

$$x - 61^\circ 52' = 70^\circ 32' \text{ or } 289^\circ 28'$$

$$x = 132^\circ 44' \text{ or } 351^\circ 20'$$

④

3(a) ways $\{a, b\} = 2$
ways $\{c, d\} = 2$

ways $\{e, f\} = 4 \cdot 2$

Total ways $= 4 \cdot 2 \times 2 = 96$

(b) $(x-1)(x-2+\sqrt{3})(x-2-\sqrt{3}) = 0$

$$(x-1)(x^2-4x+1) = 0$$

$$(x-1)(x^2-4x+1) = 0$$

$$x^3-5x^2+5x-1 = 0$$

(c) (i) $y = \frac{\sin(x+7)}{\cos(x-9)}$

$$\frac{dy}{dx} = \frac{\cos(x-9)\cos(x+7) - \sin(x+7) \cdot \sin(x-9)}{\cos^2(x-9)}$$

$$= \frac{\cos(x-9-(x-7))}{\cos^2(x-9)}$$

$$= \frac{\cos 16}{\cos^2(x-9)}$$

(ii) $y = \log_x 3$

$$= \frac{\ln 3}{\ln x}$$

$$\frac{dy}{dx} = -\frac{1}{x} \cdot \frac{\ln 3}{\ln^2 x}$$

$$= -\frac{1}{x} \cdot \frac{\ln 3}{\ln x} \cdot \frac{1}{\ln x}$$

$$= -\frac{\log_x 3}{x \ln x}$$

3(d) $x = 2p$ $y = p^2$
 $\frac{dx}{dp} = 2$ $\frac{dy}{dp} = 2p$
 $\frac{dy}{dx} = \frac{\frac{dy}{dp}}{\frac{dx}{dp}} = \frac{2p}{2} = p$
 $= 1$

$m_N = -\frac{1}{p}$

Eqn normal

$y - p^2 = -\frac{1}{p}(x - 2p)$

$py - p^3 = -x + 2p$

$x + py = 2p + p^3$ (1)

$x + py = 2p + p^3$

$y(p - 2) = 2(p - 2) + p^3 - p^2$
 $= 2(p - 2) + (p - 2)(p^2 + p + 2)$
 $= (p - 2)[2 + p^2 + 4 + p^2]$

$y = p^2 + p^2 + 6$ $p \neq 2$

$\therefore x = 2p + p^3 - p^3 - p^2 - 6p$

$= -4p - 12p$

$= -4p - 4p$

$= -4(p + 2)$

$N \equiv (-4(p + 2), p^2 + p^2 + 6)$

(5)

(iii) $x = -4(p + 2)$

$x^2 = 16[p^2 + 4 + 2p \cdot 2]$

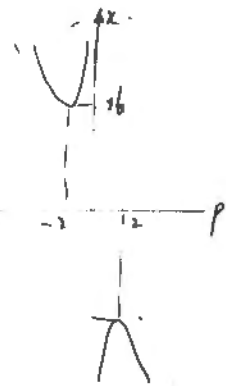
$= 16[p^2 + 4 + 4p]$

$= 16[y - 6 + 8]$

$x^2 = 16(y + 2)$

Restriction $x = -4(p + \frac{4}{p})$

Restriction $x \geq 16$ or $x \leq -16$.



4(i) $AD = \frac{h}{\tan 6^\circ}$ $BC = \frac{h}{\tan 8^\circ}$

$AC^2 = AB^2 + BC^2 - 2AB \cdot BC \cos \angle ABC$

$30^2 = \frac{h^2}{\tan^2 10^\circ} + \frac{h^2}{\tan^2 8^\circ} - \frac{2h^2 \cos 10^\circ}{\tan 10^\circ \tan 8^\circ}$

$= h^2 \left[\frac{\tan^2 8^\circ + \tan^2 10^\circ - 2 \tan 8^\circ \tan 10^\circ \cos 10^\circ}{\tan^2 10^\circ \tan^2 8^\circ} \right]$

$h = \frac{30 \tan 8^\circ \tan 10^\circ}{\sqrt{\tan^2 8^\circ + \tan^2 10^\circ - 2 \tan 8^\circ \tan 10^\circ \cos 10^\circ}}$

$= 3.05 \text{ m.}$

(2)

(-2)

(6)

(a) $\angle DAC = \angle ABC$ (Angle between chord & tangent)
 $= 2$

$\therefore \angle DCA = 180^\circ - 2 - (90^\circ - 2)$ (Angle sum $\triangle ADC$)
 $= 90^\circ + 2$

$\therefore \angle BAC = 90^\circ$ (Exterior angle $\triangle ABC$)

$\therefore BC$ is diameter (angle semi circle)

(b)(i) $y = \frac{x^2 - 3x + 2}{x^2 + 4x + 3}$

$y' = \frac{(x^2 + 4x + 3)(2x - 3) - (x^2 - 3x + 2)(2x + 4)}{(x^2 + 4x + 3)^2}$

$= \frac{2x^3 - 3x^2 + 8x^2 - 12x + 6x - 9 - [2x^3 + 4x^2 - 6x^2 - 12x + 8x + 8]}{(x^2 + 4x + 3)^2}$

$= \frac{+5x^2 - 6x - 9 - [-2x^2 - 8x + 8]}{(x^2 + 4x + 3)^2}$

$= \frac{7x^2 + 2x - 17}{(x^2 + 4x + 3)^2}$

Turning points $y' = 0$

$7x^2 + 2x - 17 = 0$

$x = \frac{-2 \pm \sqrt{4 + 4 \times 7 \times 17}}{14}$

$= \frac{-2 \pm \sqrt{120}}{7}$

$= \frac{-1 \pm 2\sqrt{30}}{7}$ $x = -1.71$ or 1.42
 $y = -10.98$ -0.023

(7)

Test
 $x = -1.71$

for nature turning points test gradient $\frac{dy}{dx}$ continuous comp $x = -1$ & $x = -3$

x	-1.8	-1.71	-1.6
$\frac{dy}{dx}$	2.26	0	-3.23

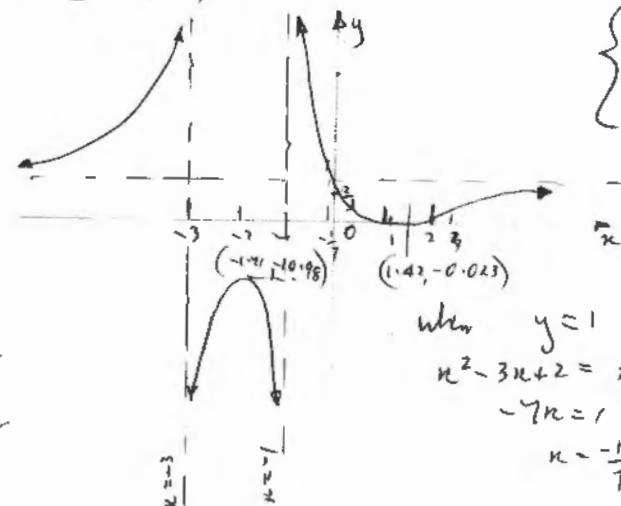
relative max at $(-1.71, -10.98)$

Test
 $x = +1.42$

x	1.3	1.42	1.5
$\frac{dy}{dx}$	-0.03	0	0.01

relative min at $(1.42, -0.023)$

(ii) $y = \frac{(x-1)(x-2)}{(x+3)(x+1)}$



- 1 - Asymptotes
- 1/2 - Roots
- 1/2 - y intercept
- 1 - Shape

when $y = 1$

$x^2 - 3x + 2 = x^2 + 4x + 3$

$-7x = 1$

$x = -\frac{1}{7}$

(8)